Testing Preprocessing on Effectiveness of CNN Detection of Tumors in Mammograms

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Background



Mammogram Image Showing Malignant Tumor

1 in 12
women diagnosed with breast cancer and
1 in 71
die because of it.

Early detection reduces mortality by
41%

Mammography is the most common method used to diagnose breast cancer.

Mammography:



Low dose X-Ray imaging modality

Application of AI:

- Increased scarcity of radiologists → automatic tumor detection and segmentation with AI being implemented to multi-stage diagnosis
- Accuracy: 85% (average radiologist) → 97.9% (with DL assistance)

Current use cases

Traditional Machine Learning Approach:

1) Feature extraction (Fourier, PCA)
2) Normalization
3) Classifier/Regressor (KNN, Random Forest)
4) Feature Selection

Deep Learning Approach:

1) Data preprocessing (normalization, NOT filtering)
2) Model architecture (CNN)
3) Optimization and data augmentation

Our question

It is uncommon for current deep learning models to perform pre-processing beyond basic normalization... does pre-processing affect model (CNN) accuracy in tumor segmentation?

Our Approach

1

Choosing dataset

- Open source
- Pre-annotated

2

Data preparation

- Dcm → PNG
- Json → binary masks

3

Pre-processing

- Median filtering
- CLAHE
- PCA

4

Model Training

- U-Net CNN architecture
- 80/20 training split
- Adam optimizer

5

Model Comparison

- Dice score
- IoU score
- Qualitative Review

Dataset

Kashiwada, Y., Takaya, E., Hiroya, M., Matsuda, N., Yashima, T., Kobayashi, T., Tamiya, G., & Ueda, T. (2025). TOMPEI-CMMD Dataset (Version 1) [Dataset]. The Cancer Imaging Archive. https://doi.org/10.7937/WEZW-BH22

Contents

- **4,162** DICOM images for **1,363** subjects
  - .dcm file is a high quality medical imaging file that contains all necessary metadata such as patient information and pixel data.
- **1,385** Image segmentation .json files

- **1,775** Chinese patients examined between July 2012 and January 2016
- An experienced radiologist diagnosed each patient with their specific lesion type and location
- 5 radiology technologists annotated each lesion's shape

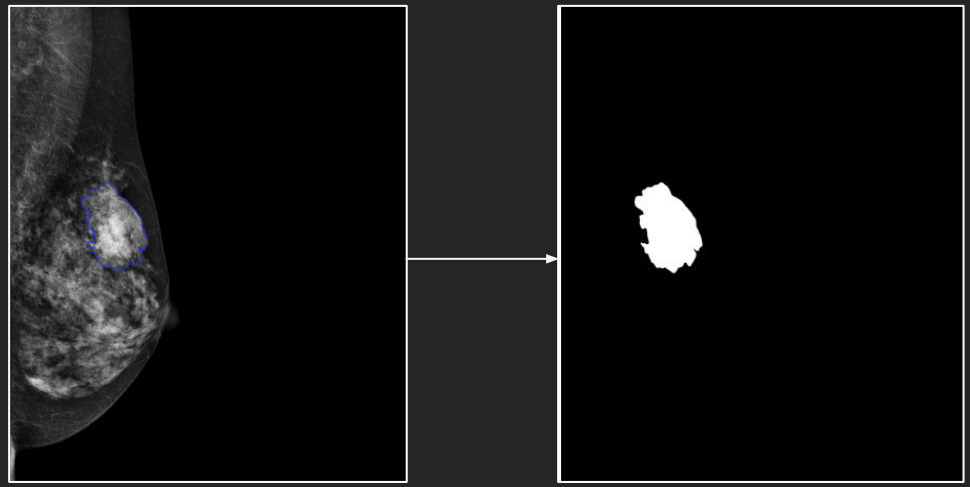
Data Preparation

Images

- The dataset includes breasts with many findings, including
  - masses
  - calcifications
  - lipomas
  - etc
- We have chosen only images with **masses** to train and locate
- We also saved each image as a .**png** for training

Binary Mask

Convert the json annotation files to image that can be used to train and verify the segmentation



Pre Processing (test 1)

Median Filtering

$$y_{(m,n)} = \text{median}\{x_{(i,j)}, (i,j) \in N\},$$

- Inspired by *Sha et al. (2020)*: achieved 96% sensitivity in mammogram segmentation with median filtering + CNN
- Median filtering **removes salt & pepper noise**

1	2	2
2	1	3
3	50	2

⇨

X	X	X
X	2	X
X	X	X

Median

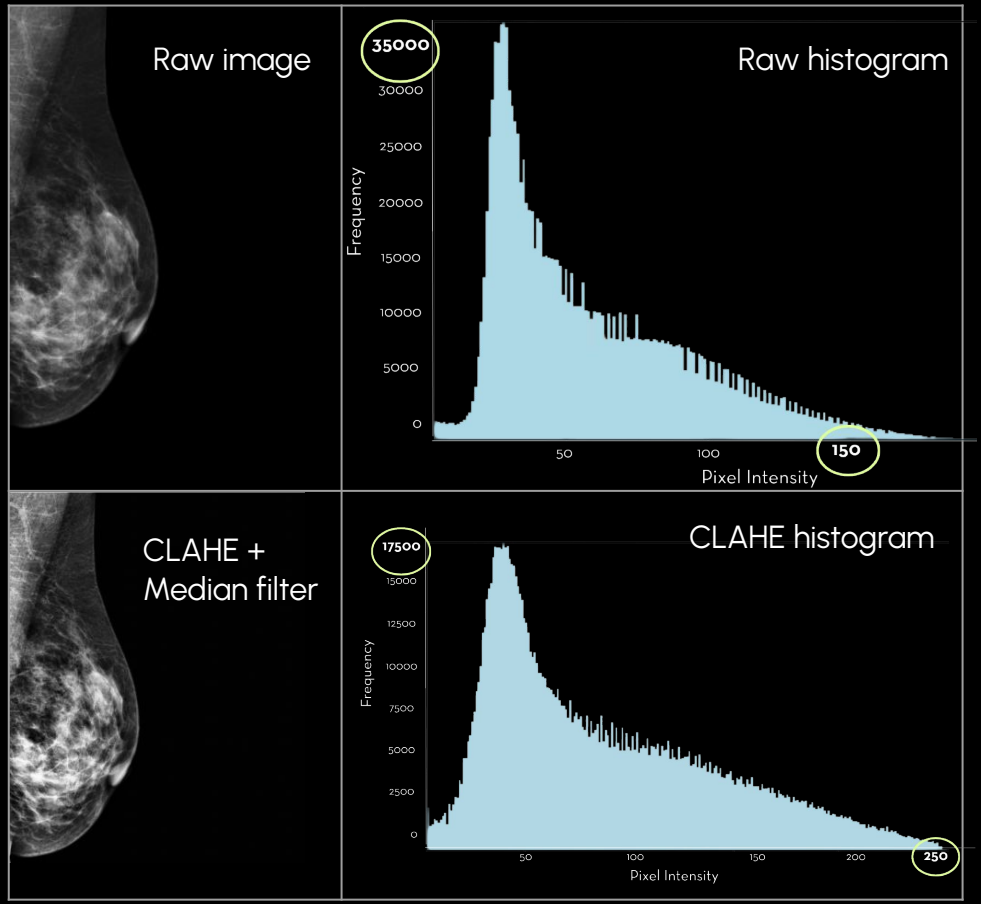
VS

X	X	X
X	7.3	X
X	X	X

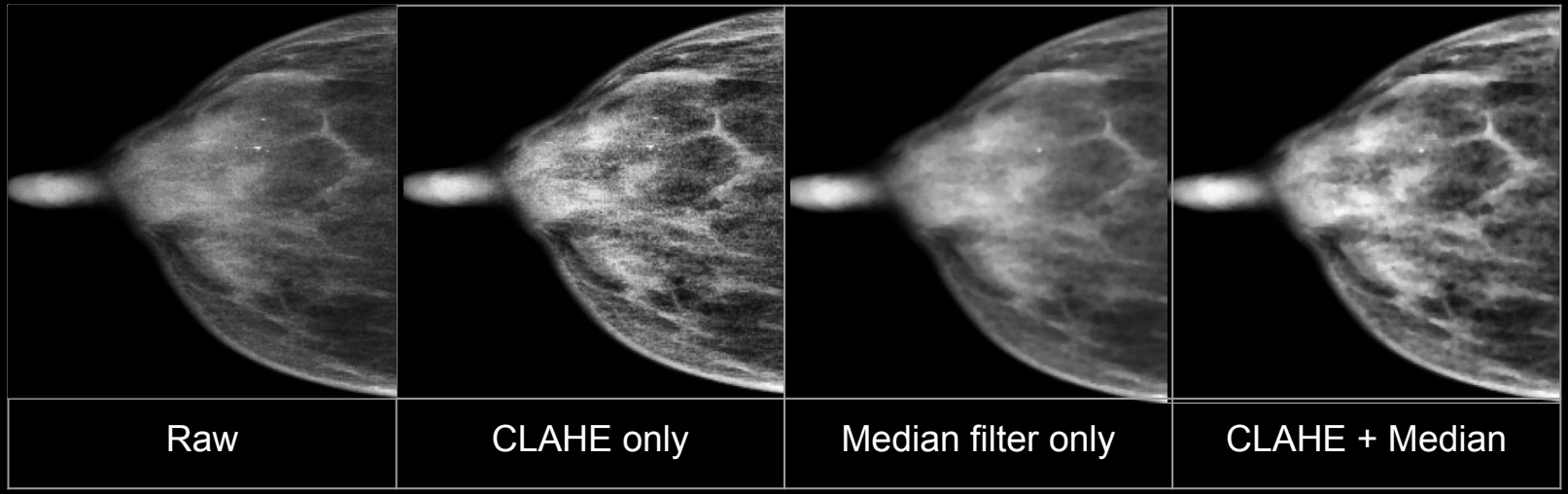
Mean

CLAHE

- Inspired by *Almajalid et al. (2018)*: CLAHE + UNet on breast ultrasound segmentation
  - DICE (overlap between masks) = .825
- Histogram equalization to distribute intensity values more evenly through whole range [0,255] → **enhance contrast**

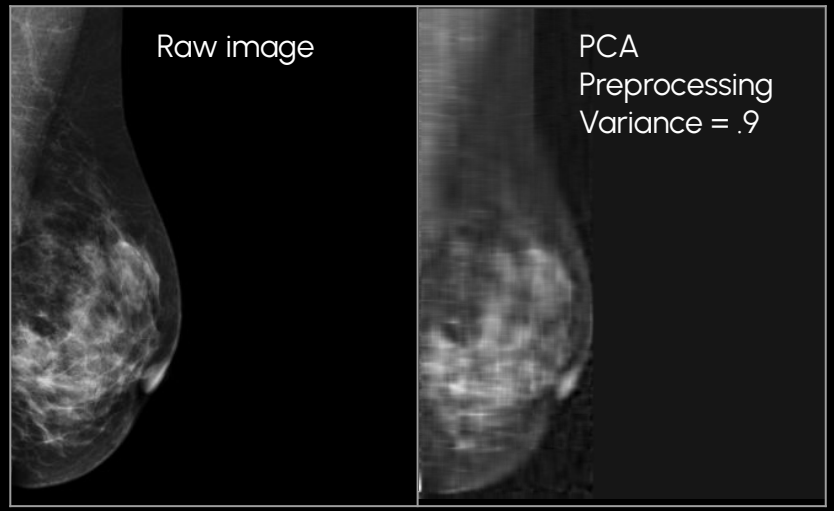


Pre Processing Comparison



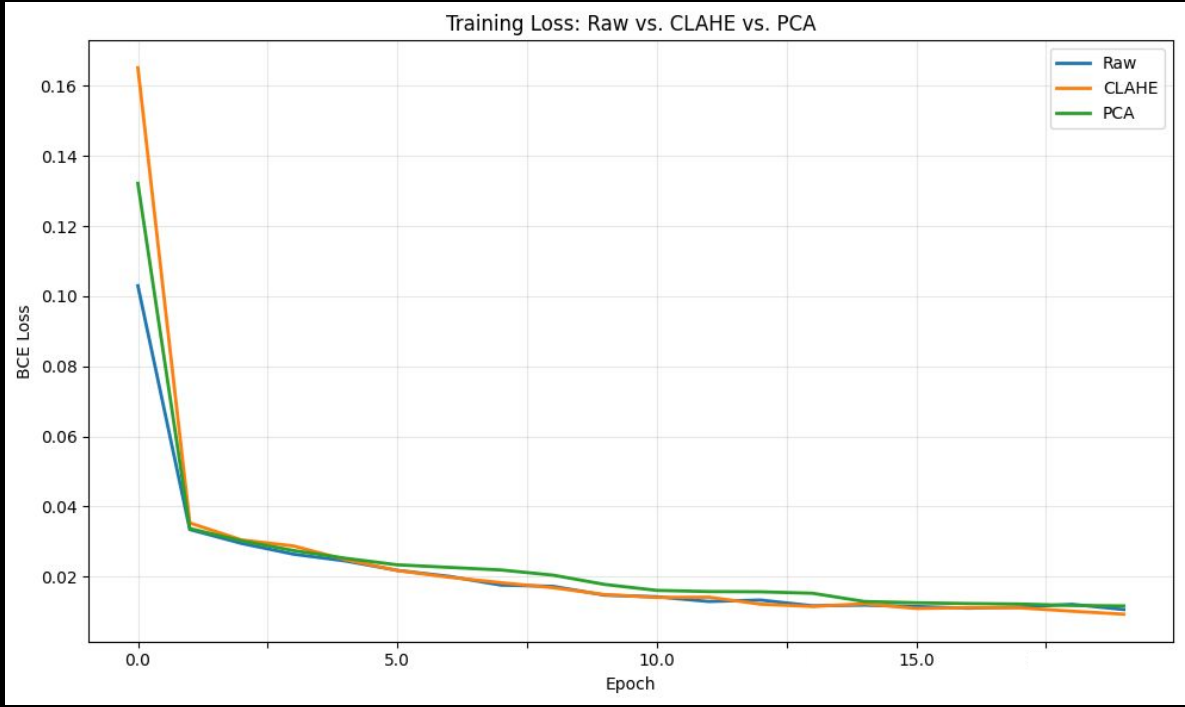
Pre Processing (test 2)

Principal Component Analysis



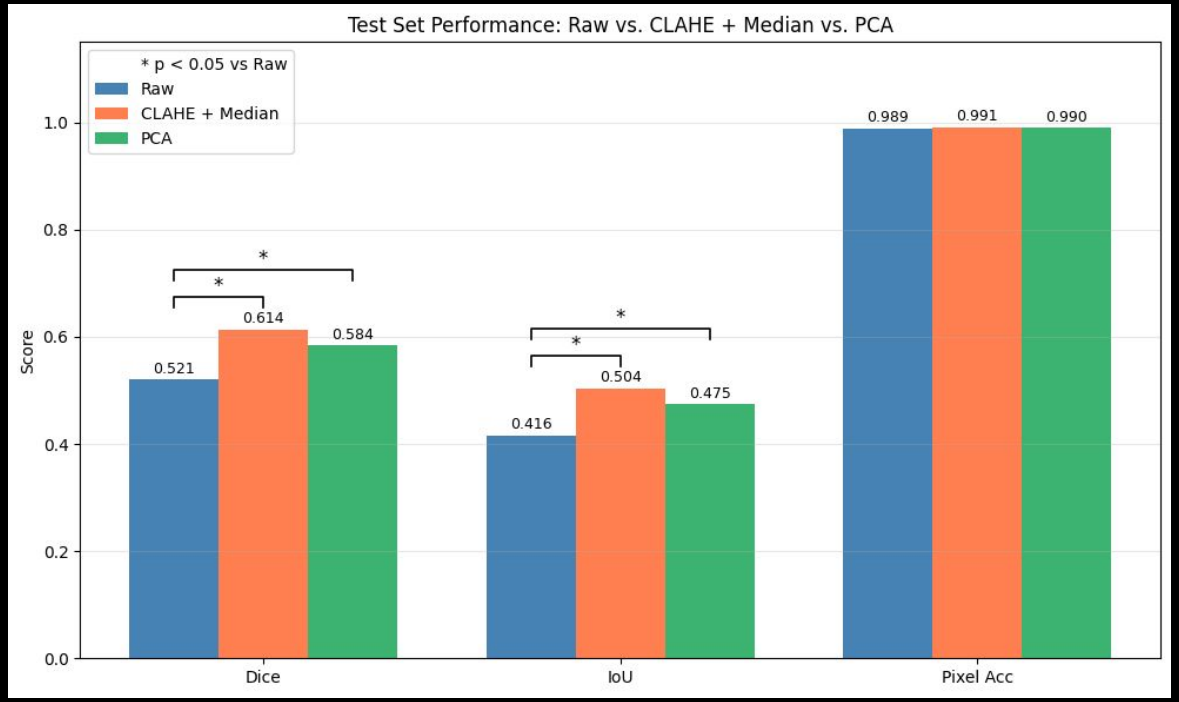
- PCA is commonly used as dimensionality reduction technique; we applied to reduce low variability
- Only keeps 90% variation (in our case), discarding the rest
- Not commonly used because potentially discarding subtle but important features like mass boundaries (low variance)

Model Training

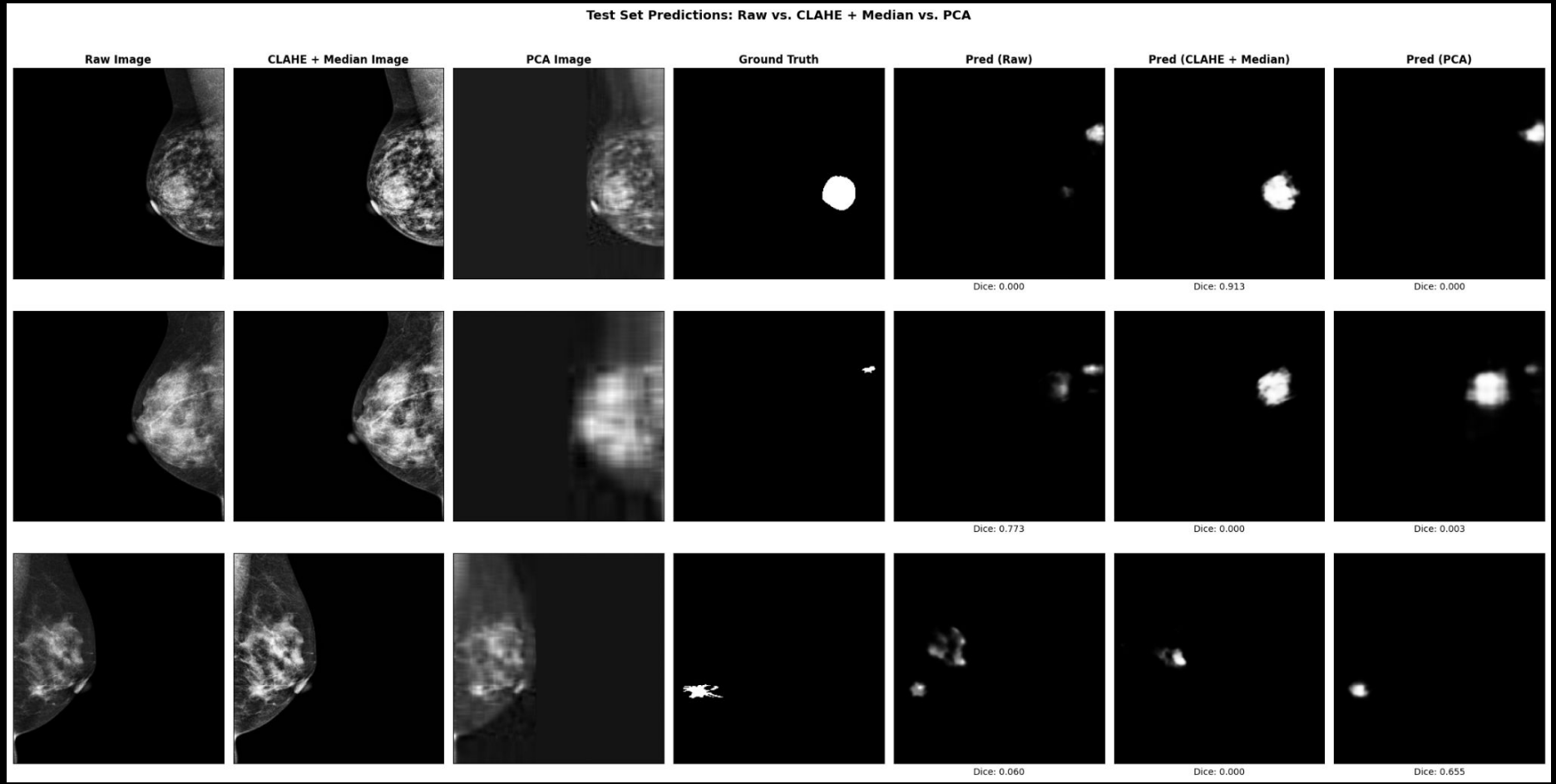


- Pre-trained starting architecture: **U-Net** with **EfficientNet-B0** encoder
- Data Split: 80% training / 20% testing split
- Training:
  - Adam optimizer (LR=1e-3)
  - Batch Size of 4, trained for 20 epochs.
- **Binary Cross-Entropy (BCE) Loss** $=-[y\log(p)+(1-y)\log(1-p)]$
  - Y = true label (0 or 1)
  - P = predicted probability (0 to 1)
  - Penalizes confident, wrong predictions

Model Comparison



- 4 options:
  - True Positive (TP)
  - True Negative (TN)
  - False Positive (FP)
  - False Negative (FN)
- **Dice score**
  - 2TP/(2TP + FP +FN)
  - Heavily weights TP, ignores TN
- **IoU score**
  - TP/(TP+FP+FN)
  - Penalizes bad classification more than DICE
- **Pixel Accuracy**
  - (TP+TN)/(TP+TN+FP+FN)
  - Sensitive to large backgrounds


Test Set Predictions: Raw vs. CLAHE + Median vs. PCA
Raw Image
CLAHE + Median Image
PCA Image
Ground Truth
Pred (Raw)
Pred (CLAHE + Median)
Pred (PCA)
Dice: 0.000
Dice: 0.913
Dice: 0.000
Dice: 0.773
Dice: 0.000
Dice: 0.003
Dice: 0.060
Dice: 0.000
Dice: 0.655

Limitations:

Data

- **Large datasets** are expensive and necessary for deep learning
- Data sharing is hard to access and largely compliant due to **HIPPA**

Unnecessary

- Preprocessing enhances images for human viewers
- Deep networks should be able to learn from original images.

Potential next steps:

Training a generalized model

Train on diverse data sets (mammograms with various sizes, angles, etc.).

Integrate Support Vector Machines to expand model to detection

CNN used for feature extraction and SVM for classification showed 98.44% accuracy [Alkhaleefah and Wu, 2019]

Thank you

Sources:
Median filtering: Sha Z, Hu L, Rouyendegh BD. Deep learning and optimization algorithms for automatic breast cancer detection. Int J Imaging Syst Technol. 2020;30:495–506. https://doi.org/10.1002/ima.22400
CLAHE: Almajalid, Rania, et al. "Development of a deep-learning-based method for breast ultrasound image segmentation." 2018 17th IEEE international conference on machine learning and applications (ICMLA). IEEE, 2018.